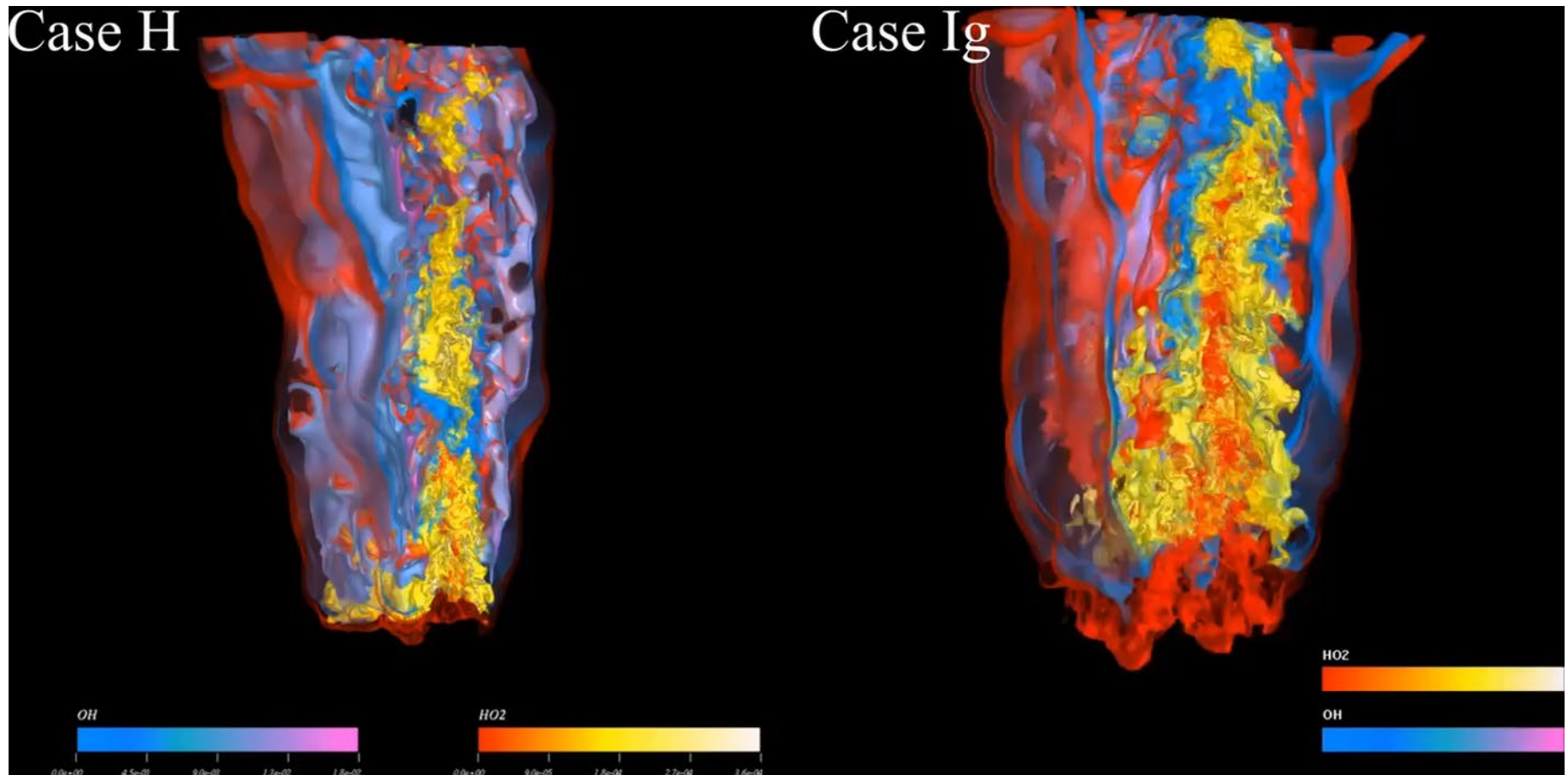


실습 가이드: OpenFOAM Reacting flow

2026 한국연소학회 가을 Tutorial
Next-Gen Combustion

Reacting flow

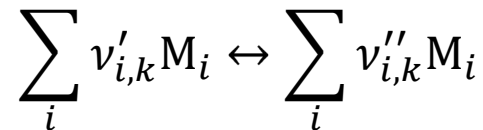
- Turbulent reacting flow [*]



Reacting flow

● Chemical reaction rates

- Species production rates are calculated from elementary reaction rates
- Consider elementary reversible reactions



- W_i is molecular weight, $c_j = \rho_j/W_j = Y_j\rho/W_j = X_j p/R^0T$

$$\dot{w}_i = W_i \hat{\omega}_i = W_i \sum_k (\nu''_{i,k} - \nu'_{i,k}) \omega_k$$

$$\omega_k = k_{k,f} \prod_j c_j^{\nu'_{j,k}} - k_{k,b} \prod_j c_j^{\nu''_{j,k}}$$

- The reaction rate constant k is determined by the modified Arrhenius law

$$k = AT^\beta \exp(-E_a/RT)$$

Reacting flow

- **Governing eqautions of multiComponentFluid**

- A **solver for transient, compressible, turbulent reacting flow** included in the standard OpenFOAM framework
- Designed to simulate reacting flows with detailed **chemistry, heat release, and turbulence** interactions

$$\text{rhoEqn: } \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0,$$

$$\text{UEqn: } \frac{\partial (\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u} \mathbf{u}) - \nabla \cdot \boldsymbol{\tau} = -\nabla p,$$

$$\text{YEqn: } \frac{\partial \rho Y_i}{\partial t} + \nabla \cdot (\rho \mathbf{u} Y_i) = \nabla \cdot (\mu_{\text{eff}} \nabla Y_i) + w_i$$

$$\text{EEqn: } \frac{\partial (\rho h)}{\partial t} + \nabla \cdot (\rho \mathbf{u} h) + \frac{\partial (\rho K)}{\partial t} + \nabla \cdot (\rho \mathbf{u} K) = \frac{\partial p}{\partial t} + \nabla \cdot (\alpha_{\text{eff}} \nabla h) + \sum h_i^0 w_i$$

————— Turbulence model

————— Combustion model

Reacting flow

● Turbulence model

- Turbulence model mathematically represents turbulent motions that **cannot be directly resolved by the computational mesh**

◆ Reynolds-Averaged Navier–Stokes (RANS)

- Turbulent fluctuations are modeled through Reynolds averaging
 - One-equation model
 - Spalart–Allmaras
 - Two-equation models
 - **$k - \varepsilon$ model**
 - RNG $k - \varepsilon$ / Realizable $k - \varepsilon$ model
 - $k - \omega$ model

◆ **$k - \varepsilon$ model**

- Solves transport equations for turbulent kinetic energy **k** and its dissipation rate **ε**
- Evaluates turbulent viscosity as $\nu_t = C_\mu k^2 / \varepsilon$
- Provides turbulence quantities used by the **EDC** and **PaSR** combustion models.

Reacting flow

● Combustion model

- Combustion model mathematically represents and simplifies **mixing, chemical reactions and heat transfer** so that they can be simulated in CFD
- Combustion modeling is classified as **Finite rate chemistry model** and **Fast chemistry model**

◆ Fast chemistry ($Da \gg 1$)

- Eddy Dissipation model (EDM)
- Equilibrium model
- Flamelet-based models (Steady Laminar Flamelet model, Flamelet Generated Manifold model, etc.)

◆ Finite rate chemistry

- Laminar finite rate model
- Eddy Dissipation model/Laminar finite rate model
- **Eddy Dissipation Concept model (EDC)**
- **Partially Stirred Reactor model (PaSR)**
- Transported PDF model

Reacting flow

● Eddy Dissipation Concept (EDC)

- Eddy Dissipation Concept (**EDC**) model is suggested by Magnussen(1981) [*]
- Based on the assumption that chemical reaction occurs in the regions where the dissipation of turbulence energy takes place

Fine structure length fraction

$$\gamma_L = C_\gamma \left(\frac{\nu \varepsilon}{k^2} \right)^{1/4} \quad C_\gamma = 2.1377$$

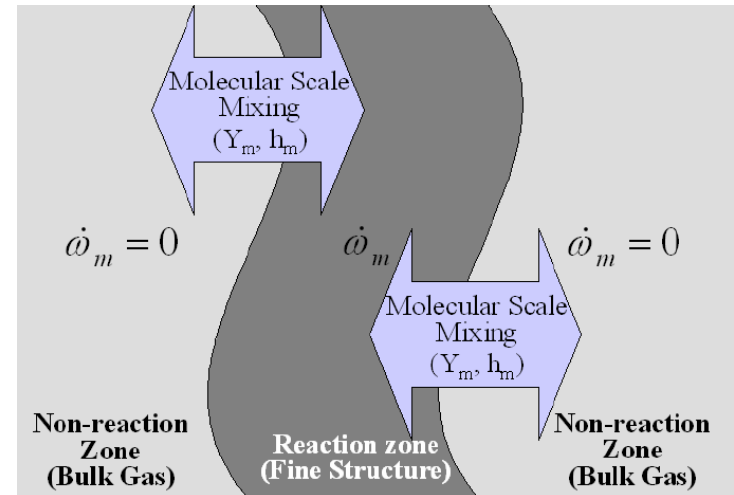
Fine structure residence time

$$\tau^* = C_\tau \left(\frac{\nu}{\varepsilon} \right)^{1/2} \quad C_\tau = 0.4083$$

Mean reaction rate

$$\bar{w}_i = \frac{\gamma_L^2}{1 - \gamma_L^2} \cdot \frac{\bar{\rho}(Y_i^* - \tilde{Y}_i)}{\tau^*}$$

 : Model coefficients



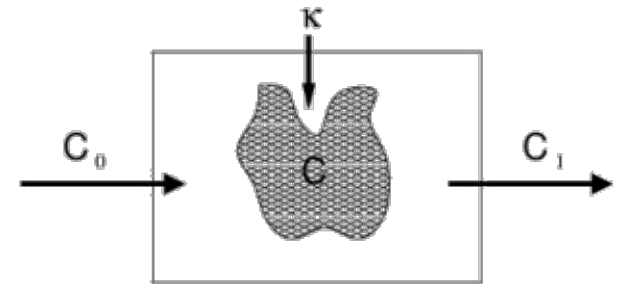
Reacting flow

● Partially-Stirred Reactor (PaSR)

- Partially stirred reactor model is suggested by Chomiak (1990) [*]
- Based on the assumption that reaction zone is perfectly stirred

Mass fraction of reaction zone

$$\kappa = \frac{\tau_c}{\tau_c + \tau_{mix}}$$



Chemical time scale & turbulent time scale

$$\tau_c = \frac{n_r Y_{tot}^*}{\sum_{n=1}^{n_r} \left(\frac{dY_{n,forward}}{dt} \cdot \nu_{n,sum} \right)} \quad \& \quad \tau_{mix} = \sqrt{\frac{k}{\varepsilon} \left(\frac{\nu}{\varepsilon} \right)^{0.5}} = \boxed{C_{mix}} \sqrt{\frac{\mu_{eff}}{\bar{\rho} \varepsilon}}$$

Mean reaction rate

$$\bar{w}_i = \kappa \cdot \frac{\bar{\rho} (Y_i^* - \tilde{Y}_i)}{\tau_{mix}}$$

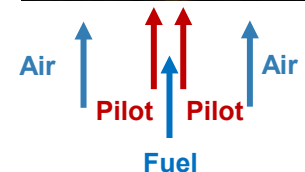
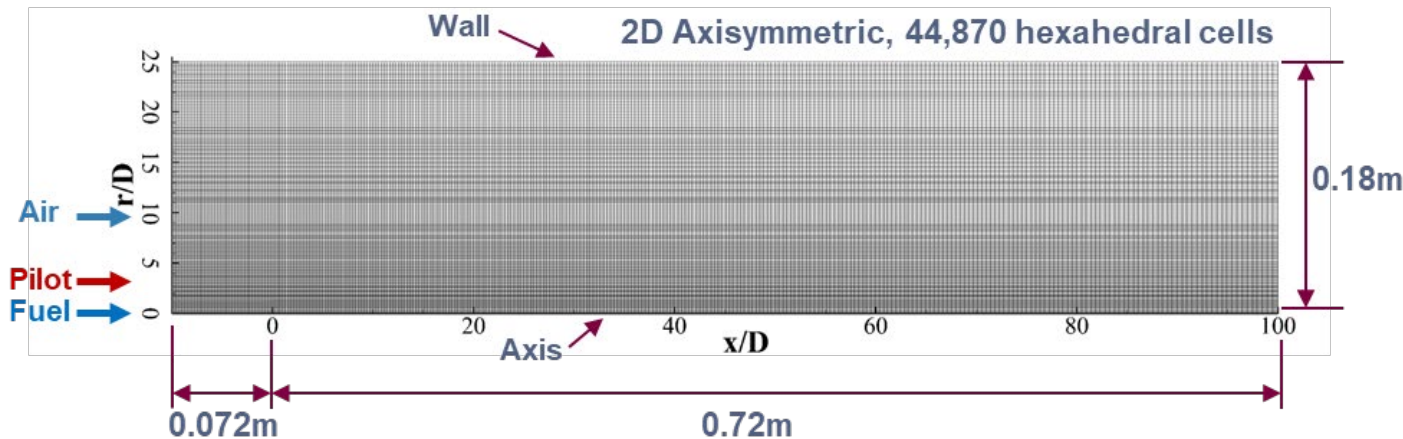
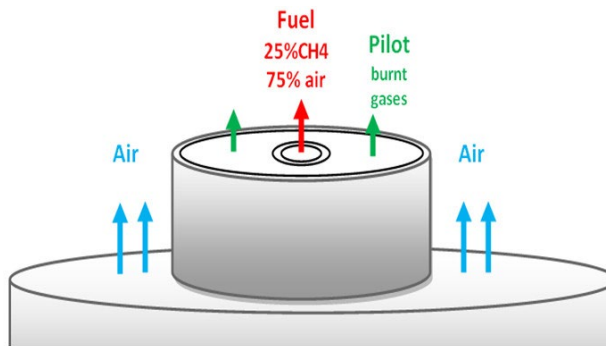
$$C_{mix} = \sqrt{\frac{1}{1 + C_\mu Re_t}}$$

Laminar flow : $C_{mix} = 1$
 Typical turbulent flow : $C_{mix} = 0.001 \sim 0.3$
 Extremely turbulent flow : $C_{mix} = 0$

Reacting flow

● Tutorial 2: Sandia Flame D [*]

- A piloted partially premixed methane/air diffusion flame
- Widely used for validation of modeling



Reacting flow

● Tutorial 2: Sandia Flame D [*]

- Fuel : CH₄ (25% by vol.) + Air (75% by vol.)
- Pilot : Lean mixture ($\phi = 0.77$)
- Coflow : Dry air

	Fuel	Pilot	Coflow air
U [m/s]	49.6	11.4	0.9
T [K]	294	1880	291
Y_{CH_4}	0.1561	0	0
Y_{CO}	0	4.070E-03	0
Y_{CO_2}	0	1.098E-01	0
Y_{H}	0	2.480E-05	0
Y_{H_2}	0	1.290E-04	0
$Y_{\text{H}_2\text{O}}$	0	9.420E-02	0
Y_{N_2}	0.6473	7.342E-01	0.77
Y_{O}	0	7.470E-04	0
Y_{O_2}	0.1966	5.400E-02	0.23
Y_{OH}	0	2.800E-03	0
Diameter [mm]	7.2	18.2	-

Reacting flow

- Preprocessing: Initial & boundary conditions

①

Command Line

```
cd $FOAM_RUN/KOSCO-2026-NextGen-Combustion/Tutorial/2.SandiaFlamed
```

```
ll 0/
```

```
ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/2.SandiaFlamed$ ll 0
total 84
drwxr-xr-x 2 ghkrk ghkrk 4096 Oct  7 15:56 ./
drwxr-xr-x 6 ghkrk ghkrk 4096 Oct  7 15:57 ../
-rw-r--r-- 1 ghkrk ghkrk 1391 Oct  7 15:56 CH4.orig
-rw-r--r-- 1 ghkrk ghkrk 1391 Oct  7 15:56 CO
-rw-r--r-- 1 ghkrk ghkrk 1391 Oct  7 15:56 CO2
-rw-r--r-- 1 ghkrk ghkrk 1047 Oct  7 15:56 G
-rw-r--r-- 1 ghkrk ghkrk 1390 Oct  7 15:56 H
-rw-r--r-- 1 ghkrk ghkrk 1391 Oct  7 15:56 H2
-rw-r--r-- 1 ghkrk ghkrk 1391 Oct  7 15:56 H2O
-rw-r--r-- 1 ghkrk ghkrk 1401 Oct  7 15:56 N2.orig
-rw-r--r-- 1 ghkrk ghkrk 1390 Oct  7 15:56 O
-rw-r--r-- 1 ghkrk ghkrk 1400 Oct  7 15:56 O2.orig
-rw-r--r-- 1 ghkrk ghkrk 1390 Oct  7 15:56 OH
-rw-r--r-- 1 ghkrk ghkrk 1406 Oct  7 15:56 T.orig
-rw-r--r-- 1 ghkrk ghkrk 1442 Oct  7 15:56 U
-rw-r--r-- 1 ghkrk ghkrk 1391 Oct  7 15:56 Ydefault
-rw-r--r-- 1 ghkrk ghkrk 1598 Oct  7 15:56 alphas
-rw-r--r-- 1 ghkrk ghkrk 1985 Oct  7 15:56 epsilon
-rw-r--r-- 1 ghkrk ghkrk 1651 Oct  7 15:56 k
-rw-r--r-- 1 ghkrk ghkrk 1680 Oct  7 15:56 nut
-rw-r--r-- 1 ghkrk ghkrk 1313 Oct  7 15:56 p
```

Additionally required for reacting flow
(initial conditions of species)

Reacting flow

● Preprocessing: Initial & boundary conditions

2 Command Line

```
vi 0/N2.orig
```

To exit vi editor, type **:q**

```
:wq  write and quit the file
:w    write
:q    quit the file
:q!   quit the file (ignore the change)
```

- Species compositions in OpenFOAM are specified using mass fractions, Y_i
- Repeat this step for every other species with a non-zero mass fraction in the initial internal field or at any inlet boundary.

```
/*----- C++ -----*/
//
// \ V /   F i e l d       O p e r a t i o n       A n d       M a n i p u l a t i o n
//  / \   / O p e r a t i o n       A n d       M a n i p u l a t i o n
//
OpenFOAM: The Open Source CFD Toolbox
Website:  https://openfoam.org
Version:  12
/*-----*/

FoamFile
{
    format      ascii;
    class       volScalarField;
    location    "0";
    object      N2;
}
// *****

dimensions      [];

internalField    uniform 0.77;

boundaryField
{
    inletCH4
    {
        type      fixedValue;
        value      uniform 0.6473;
    }

    wallOutside
    {
        type      zeroGradient;
    }

    wallTube
    {
        type      zeroGradient;
    }

    inletPilot
    {
        type      fixedValue;
        value      uniform 0.7342;
    }

    inletAir
    {
        type      fixedValue;
        value      uniform 0.77;
    }
}
```

Reacting flow

● Preprocessing: Initial & boundary conditions

3

Command Line

```
vi 0/Ydefault
```

- Ydefault consisted of zero in initial internal field and all inlet boundary fields.
- When all species contained in kinetic mechanism are not present in the current directory, it serves as a substitute for the remaining absent species.

```
/*----- C++ -----*/
//
// \ V / F i e l d
//  \ / O p e r a t i o n
//   V A n d
//   M a n i p u l a t i o n
//
// OpenFOAM: The Open Source CFD Toolbox
// Website: https://openfoam.org
// Version: 12
//-----*/

FoamFile
{
    format      ascii;
    class       volScalarField;
    location    "0";
    object      Ydefault;
}
// ***** //

dimensions      [];

internalField    uniform 0;

boundaryField
{
    wallTube
    {
        type      zeroGradient;
    }

    outlet
    {
        type      inletOutlet;
        inletValue $internalField;
    }

    inletPilot
    {
        type      fixedValue;
        value      uniform 0;
    }

    inletAir
    {
        type      fixedValue;
        value      uniform 0;
    }

    wallOutside
    {
        type      zeroGradient;
    }
}
```

Reacting flow

● Kinetic mechanism

- Chemkin is a software for solving chemical kinetic problems
- Its format is widely used in many simulations of reacting flows
- 1) Gas-phase chemistry file (chem.inp)
 - Elements, Gas phase species, Gas-phase reactions
 - Include the values of A , β , E_a for modified Arrhenius rate coefficient
- 2) Thermodynamic data file (therm.dat)
 - Coefficients of polynomial fits to temperature for species specific heats, enthalpy, and entropy
- 3) Transport data file (tran.dat)
 - Fundamental molecular parameters
 - Lennard-Jones well depth, Lennard-Jones diameter, dipole moment, polarizability, rotational relaxation number
- Example of kinetic mechanism for natural gas combustion – GRI 3.0
 - <http://combustion.berkeley.edu/gri-mech/version30/text30.html>

Reacting flow

● Preprocessing: Kinetic mechanism

- Chemkin-format Gas-phase chemistry file

4

Command Line

```
vi chemkin/chem.inp
```

Elements

Species

Reactions

```
! GRI-Mech Version 3.0 7/30/99 CHEMKIN-II format
! See README30 file at anonymous FTP site unix.sri.com, directory gri;
! WorldWideWeb home page http://www.me.berkeley.edu/gri_mech/ or
! through http://www.gri.org , under 'Basic Research',
! for additional information, contacts, and disclaimer

ELEMENTS
O H C N AR
END

SPECIES
H2      H      O      O2      OH      H2O      HO2      H2O2
C       CH     CH2     CH2(S)  CH3     CH4      CO       CO2
HCO     CH2O    CH2OH   CH3O    CH3OH   C2H      C2H2     C2H3
C2H4    C2H5     C2H6    HCCO    CH2CO   HCCOH    N2       AR
C3H7    C3H8     CH2CHO   CH3CHO
END

REACTIONS
2O+M<=>O2+M 1.200E+17 -1.000 .00
H2/ 2.40/ H2O/15.40/ CH4/ 2.00/ CO/ 1.75/ CO2/ 3.60/ C2H6/ 3.00/ AR/ .83/
O+H+M<=>OH+M 5.000E+17 -1.000 .00
H2/2.00/ H2O/6.00/ CH4/2.00/ CO/1.50/ CO2/2.00/ C2H6/3.00/ AR/ .70/
O+H2<=>H+OH 3.870E+04 2.700 6260.00
O+H2<=>OH+O2 2.000E+13 .000 .00
O+H2O2<=>OH+HO2 9.630E+06 2.000 4000.00
O+CH<=>H+CO 5.700E+13 .000 .00
O+CH2<=>H+HCO 8.000E+13 .000 .00
O+CH2(S)<=>H2+CO 1.500E+13 .000 .00
O+CH2(S)<=>H+HCO 1.500E+13 .000 .00
O+CH3<=>H+CH2O 5.060E+13 .000 .00
O+CH4<=>OH+CH3 1.020E+09 1.500 8600.00
O+CO(+M)<=>CO2(+M) 1.800E+10 .000 2385.00
```

Reacting flow

- Preprocessing: Kinetic mechanism

- Chemkin-format Gas-phase thermodynamic data file

5

Command Line

```
vi chemkin/therm.dat
```

```
THERMO ALL
  250.000  1000.000  5000.000
! GRI-Mech Version 3.0 Thermodynamics released 7/30/99
! NASA Polynomial format for CHEMKIN-II
! see README file for disclaimer
0          L 1/900  1          G  200.000  3500.000  1000.000  1
  2.56942078E+00-8.59741137E-05 4.19484589E-08-1.00177799E-11 1.22833691E-15 2
  2.92175791E+04 4.78433864E+00 3.16826710E+00-3.27931884E-03 6.64306396E-06 3
-6.12806624E-09 2.11265971E-12 2.91222592E+04 2.05193346E+00 4
```

Polynomial Coefficients for NASA7

Reacting flow

● Preprocessing: Kinetic mechanism

- The standard multicomponent solver module uses the Sutherland transport model to calculate viscosity (μ_i) and thermal conductivity (λ_i).

6

Command Line

```
vi chemkin/transportSutherland
```

$$\mu_i = \frac{A_s \sqrt{T}}{1 + T_s/T}$$

$$\lambda_i = \mu_i c_v \left(1.32 + 1.77 \frac{R}{c_v} \right)$$

```
/*----- C++ -----*/
//
//      F ield      | OpenFOAM: The Open Source CFD Toolbox
//      O peration   | Website: https://openfoam.org
//      A nd         | Version: 12
//      M anipulation|
//
FoamFile
{
    format      ascii;
    class       dictionary;
    location    "chemkin";
    object      transportSutherland;
}
// ***** //

"."
{
    transport
    {
        As 1.512e-06;
        Ts 120.;
    }
}
```

Reacting flow

● Preprocessing: Kinetic mechanism

- Convert Chemkin-format files into OF-format reaction and thermodynamic file

⑦

Command Line

```
chemkinToFoam \  
chemkin/chem.inp \  
chemkin/therm.dat \  
chemkin/transportSutherland \  
constant/reactions \  
constant/thermo.STM
```

Input

Output

```
vi constant/reactions
```

```
reactions  
{  
  un-named-reaction-0  
  {  
    type          reversibleThirdBodyArrhenius;  
    reaction      "2O = O2";  
    A             1.2e+11;  
    beta          -1;  
    Ta            0;  
    coeffs
```

Reaction rate type
Reaction
Arrhenius coefficients

Reacting flow

- **Preprocessing: Kinetic mechanism**

- Convert Chemkin-format files into OF-format reaction and thermodynamic file

8

Command Line

```
vi constant/thermo.STM
```

Species list

```
species 36 ( H2 H O O2 OH H2O H02 H2O2 C CH CH2 CH2(S) CH3 CH4 CO C02 HCO CH2O CH2OH CH3O CH3OH C2H C2H2 C2H3 C2H4 C2H5 C2H6 HCCO CH2CO HCCOH N2 AR C3H7 C3H8 CH2CHO CH3CHO );
```

```
OH  
{
```

```
  specie
```

```
  {  
    molWeight 17.00737;  
  }
```

Molecular weight

```
  thermodynamics
```

```
  {  
    Tlow 200;  
    Thigh 3500;  
    Tcommon 1000;  
    highCpCoeffs ( 3.09288767 0.000548429716 1.26505228e-07 -8.79461556e-11 1.17412376e-14 3858.657 4.4766961 );  
    lowCpCoeffs ( 3.99201543 -0.00240131752 4.61793841e-06 -3.88113333e-09 1.3641147e-12 3615.08056 -0.103925458 );  
  }
```

**NASA polynomial
coefficient**

```
  transport
```

```
  {  
    As 1.512e-06;  
    Ts 120;  
  }
```

Sutherland coefficients

```
  elements
```

```
  {  
    O 1;  
    H 1;  
  }
```

```
}
```

Reacting flow

- **Preprocessing: Turbulence model**

9

Command Line

```
vi constant/momentumTransport
```

```

/*-----*- C++ -*-----*/
    \
    //=====
    //      F ield
    //      O peration
    //      A nd
    //      M anipulation
    \
    OpenFOAM: The Open Source CFD Toolbox
    Website:  https://openfoam.org
    Version:  12
    \
/*-----*/

FoamFile
{
    format      ascii;
    class       dictionary;
    location    "constant";
    object      momentumTransport;
}

// *****

simulationType  RAS;

RAS
{
    model          kEpsilon;

    turbulence      on;
    printCoeffs     on;
}

// *****

```

Reacting flow

- Preprocessing: Chemistry model

10

Command Line

```
vi constant/chemistryProperties
```

```
/*----- C++ -----*\
=====
\ \ \ F i e l d           | OpenFOAM: The Open Source CFD Toolbox
 \ \ \ O peration        | Website:  https://openfoam.org
  \ \ \ A nd              | Version:  12
   \ \ \ M anipulation    |
/*-----*\

FoamFile
{
    format      ascii;
    class       dictionary;
    location    "constant";
    object      chemistryProperties;
}
// *****

#includeEtc "caseDicts/solvers/chemistry/TDAC/chemistryPropertiesFlame.cfg"

chemistryType
{
    solver      ode;
}

chemistry      on;

initialChemicalTimeStep 1e-7;

odeCoeffs
{
    solver      seulex;
    absTol      1e-8;
    relTol      1e-1;
}
```

Reacting flow

- Preprocessing: Combustion model

11

Command Line

```
vi constant/combustionProperties
```

```
/*----- C++ -----*/
=====
\ \ \ \ \ F i e l d
\ \ \ \ \ O p e r a t i o n
\ \ \ \ \ A n d
\ \ \ \ \ M a n i p u l a t i o n
=====

FoamFile
{
    format      ascii;
    class       dictionary;
    location    "constant";
    object      combustionProperties;
}
// *****

combustionModel EDC; // For EDC
//combustionModel PaSR; // For PaSR

EDCCoeffs
{
    version v2005; // EDC model version
}

PaSRCoeffs
{
    Cmix 0.1; // Mixing time-scale coefficient
    integrateReactionRate true; // Integrate chemical reaction
}
// *****
```

For EDC

For PaSR

● Preprocessing: Mesh Generation & setFields

Command Line

setFields

```

ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/2.SandiaFlamed$ blockMesh
/*-----*/
      \   /
     //   \   F ield
    //      \ O peration
   //         \ A nd
  //            \ M anipulation
 /*-----*/
OpenFOAM: The Open Source CFD Toolbox
Website:  https://openfoam.org
Version:  12

Build : 12-86e126a7bc4d
Exec  : blockMesh
Date  : Oct 06 2026
Time  : 10:38:40
Host  : "20225263-p01"
PID   : 72615
I/O   : uncollated
Case  : /home/ghkrk/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/2.SandiaFlamed
nProcs : 1
sigFpe : Enabling floating point exception trapping (FOAM_SIGFPE).
fileModificationChecking : Monitoring run-time modified files using timeStampMaster (fileModificationSkew 10)
allowSystemOperations : Allowing user-supplied system call operations

// *****

Create time

Deleting polyMesh directory
"/home/ghkrk/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/2.SandiaFlamed/constant/polyMesh"
Reading "blockMeshDict"

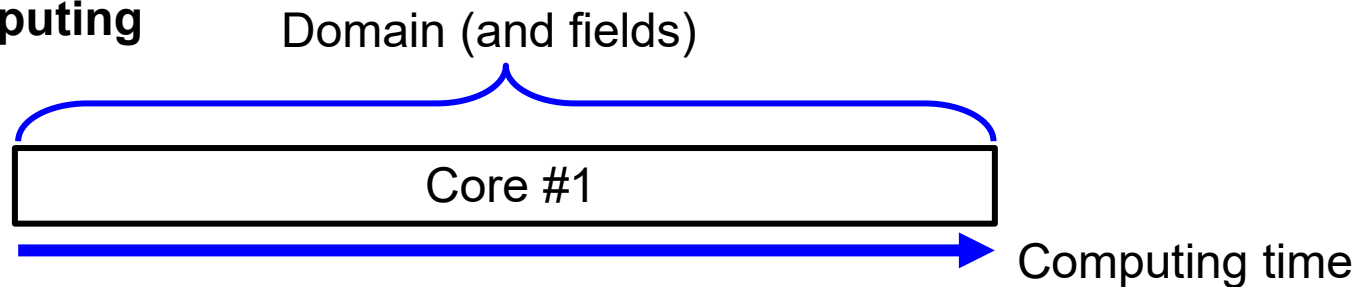
Creating block mesh from
"system/blockMeshDict"
```

Reacting flow

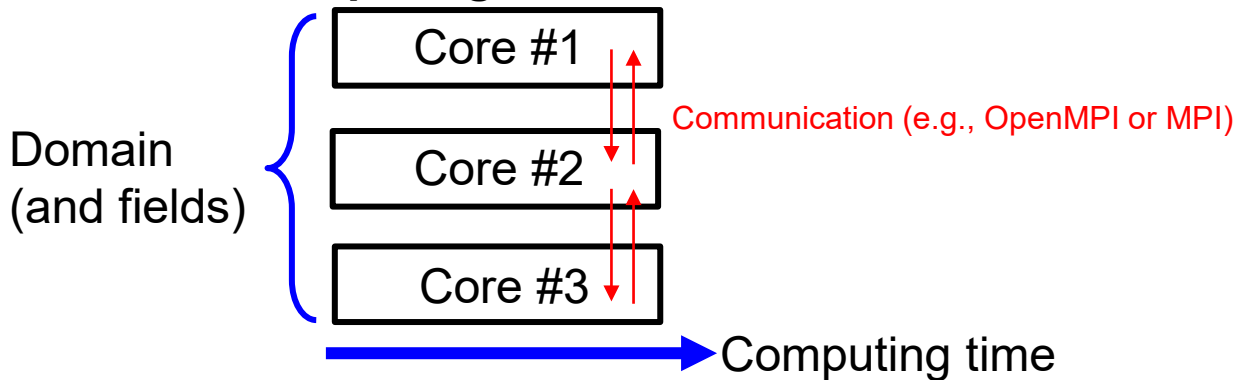
- **Preprocessing: Domain decomposition**

- Parallel computing

1. Serial computing



2. Parallel computing



Reacting flow

- Preprocessing: Domain decomposition
 - Parallel computing

In system/decomposeParDict file

```
/*-----* C++ *-----*/
//
// \ \ \ \ \
//  F i e l d
//  O p e r a t i o n
//  A n d
//  M a n i p u l a t i o n
//
// OpenFOAM: The Open Source CFD Toolbox
// Website:  https://openfoam.org
// Version:  12
//
FoamFile
{
    format      ascii;
    class       dictionary;
    note        "mesh decomposition control dictionary";
    object      decomposeParDict;
}
// ***** //

numberOfSubdomains 4;

method            simple;

simpleCoeffs
{
    n              (1 1 4);
    delta          0.001;
}

// ***** //
```

13

Command Line

```
vi system/decomposeParDict
```

Reacting flow

- Preprocessing: Domain decomposition

- Parallel computing

14

Command Line

decomposePar

11

```
Decomposing volVectorFields
```

```
U
```

```
Decomposing point fields
```

```
(no point fields)
```

```
End
```

```
ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/2.SandiaFlamed$ ll
total 48
drwxr-xr-x 10 ghkrk ghkrk 4096 Oct  6 10:29 ./
drwxr-xr-x  6 ghkrk ghkrk 4096 Oct  5 16:31 ../
drwxr-xr-x  2 ghkrk ghkrk 4096 Oct  5 16:31 0/
-rwxr-xr-x  1 ghkrk ghkrk  328 Oct  5 16:31 Allclean*
-rwxr-xr-x  1 ghkrk ghkrk  578 Oct  5 16:31 Allrun*
drwxr-xr-x  2 ghkrk ghkrk 4096 Oct  5 16:31 chemkin/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:29 constant/
drwxr-xr-x  4 ghkrk ghkrk 4096 Oct  6 10:29 processor0/
drwxr-xr-x  4 ghkrk ghkrk 4096 Oct  6 10:29 processor1/
drwxr-xr-x  4 ghkrk ghkrk 4096 Oct  6 10:29 processor2/
drwxr-xr-x  4 ghkrk ghkrk 4096 Oct  6 10:29 processor3/
drwxr-xr-x  2 ghkrk ghkrk 4096 Oct  6 10:29 system/
```

Domain is decomposed

Reacting flow

- Solving: Run the simulation
 - Use multicomponentFluid solver

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Command Line

```
mpirun -np 4 foamRun -parallel
```

Number of processors to use
(number of decomposed domain)

Command want to execute

```
ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/2.SandiaFlameD$ cd processor0
ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/2.SandiaFlameD/processor0$ ll
total 52
drwxr-xr-x 13 ghkrk ghkrk 4096 Oct  6 10:49 ./
drwxr-xr-x 18 ghkrk ghkrk 4096 Oct  6 10:49 ../
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:37 0/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:38 1000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:38 2000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:39 3000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:40 4000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:41 5000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:42 6000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:48 7000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:49 8000/
drwxr-xr-x  2 ghkrk ghkrk 4096 Oct  6 10:37 TDAC/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:37 constant/
```

Results is in each processor directory

Reacting flow

● Post-Processing

```
U
Reconstructing surfaceScalarFields

phi
Reconstructing point fields

(no point fields)

Collecting uniform files

End

ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/2.SandiaFlamed$ ll
total 92
drwxr-xr-x 18 ghkrk ghkrk 4096 Oct  6 10:49 ./
drwxr-xr-x  6 ghkrk ghkrk 4096 Oct  5 16:31 ../
drwxr-xr-x  2 ghkrk ghkrk 4096 Oct  5 16:31 0/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:43 1000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:43 2000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:43 3000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:43 4000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:43 5000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:43 6000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:49 7000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:49 8000/
-rwxr-xr-x  1 ghkrk ghkrk  310 Oct  6 10:43 Allclean*
-rwxr-xr-x  1 ghkrk ghkrk  552 Oct  6 10:42 Allrun*
drwxr-xr-x  2 ghkrk ghkrk 4096 Oct  6 10:46 chemkin/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:43 constant/
-rw-r--r--  1 ghkrk ghkrk 3342 Oct  6 10:37 log.blockMesh
-rw-r--r--  1 ghkrk ghkrk 1107 Oct  6 10:37 log.chemkinToFoam
-rw-r--r--  1 ghkrk ghkrk 1514 Oct  6 10:37 log.setFields
drwxr-xr-x 13 ghkrk ghkrk 4096 Oct  6 10:49 processor0/
drwxr-xr-x 13 ghkrk ghkrk 4096 Oct  6 10:49 processor1/
drwxr-xr-x 13 ghkrk ghkrk 4096 Oct  6 10:49 processor2/
drwxr-xr-x 13 ghkrk ghkrk 4096 Oct  6 10:49 processor3/
drwxr-xr-x  2 ghkrk ghkrk 4096 Oct  6 10:47 system/
```

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Command Line

reconstructPar

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Results of each processor is merged

Reacting flow

● Post-Processing

```
U
Reconstructing surfaceScalarFields

phi
Reconstructing point fields

(no point fields)

Collecting uniform files

End

ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/2.SandiaFlameD$ ll
total 92
drwxr-xr-x 18 ghkrk ghkrk 4096 Oct  6 10:49 ./
drwxr-xr-x  6 ghkrk ghkrk 4096 Oct  5 16:31 ../
drwxr-xr-x  2 ghkrk ghkrk 4096 Oct  5 16:31 0/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:43 1000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:43 2000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:43 3000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:43 4000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:43 5000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:43 6000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:49 7000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:49 8000/
-rwxr-xr-x  1 ghkrk ghkrk  310 Oct  6 10:43 Allclean*
-rwxr-xr-x  1 ghkrk ghkrk  552 Oct  6 10:42 Allrun*
drwxr-xr-x  2 ghkrk ghkrk 4096 Oct  6 10:46 chemkin/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:43 constant/
-rw-r--r--  1 ghkrk ghkrk 3342 Oct  6 10:37 log.blockMesh
-rw-r--r--  1 ghkrk ghkrk 1107 Oct  6 10:37 log.chemkinToFoam
-rw-r--r--  1 ghkrk ghkrk 1514 Oct  6 10:37 log.setFields
drwxr-xr-x 13 ghkrk ghkrk 4096 Oct  6 10:49 processor0/
drwxr-xr-x 13 ghkrk ghkrk 4096 Oct  6 10:49 processor1/
drwxr-xr-x 13 ghkrk ghkrk 4096 Oct  6 10:49 processor2/
drwxr-xr-x 13 ghkrk ghkrk 4096 Oct  6 10:49 processor3/
drwxr-xr-x  2 ghkrk ghkrk 4096 Oct  6 10:47 system/
```

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Command Line

```
rm -rf processor*
```

Do not need files in processor* after reconstruct

Reacting flow

● Post-Processing

- Do the same process you have done in cavity Tutorial

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Command Line

```
touch reacting.foam
```

Copy & paste files
to C: or D: drive

Open foam file
with paraview

